

Data Visualisation: Empowering Business with Effective Insights

Online Retail Store — Revenue, Customer & Demand Analysis for the CEO and CMO

Prepared by Mitali Aswale

Data Visualization Project | Tableau Analysis

About the Dataset

A transaction-level dataset from a UK-based online retailer, covering one full year of sales activity.

8

Columns

541,909

Raw Transactions

2010–2011

Time Period

Online Retail

Business Type

The 8 Columns in the Raw Data

- **InvoiceNo** Unique ID for each transaction
- **InvoiceDate** Date and time of the transaction
- **StockCode** Unique ID for each product
- **UnitPrice** Price per unit, in \$
- **Description** Product name
- **CustomerID** Unique ID for the customer
- **Quantity** Units sold in that transaction
- **Country** Customer's country

Cleaning the Data

Cleaning was applied per business question, not as one blanket rule, to avoid unnecessary data loss.

1

Quantity Check

Removed every record where Quantity was below 1 unit — these were returns, cancellations, or data entry errors that would distort revenue and demand figures.

2

Unit Price Check

Removed every record where Unit Price was below \$0 — a negative price isn't possible in reality and signals a data error.

3

Customer ID

Kept blank Customer IDs for country- and time-based charts (Q1, Q2, Q4). Excluded them only for the customer-level chart (Q3), since revenue can't be linked to an unknown customer.

This question-first approach to cleaning ensures each chart uses the maximum valid data relevant to it, rather than discarding data that other charts don't even need.

Q1

Monthly Revenue Trend — 2011

Asked by: CEO



The trend of sum of Revenue for Invoice Date Month. The marks are labeled by sum of Revenue. The data is filtered on Invoice Date Year, which ranges from 2011 to 2011.

THE QUESTION

Show the 2011 revenue trend at a monthly level, to understand seasonal patterns and support next year's forecast.

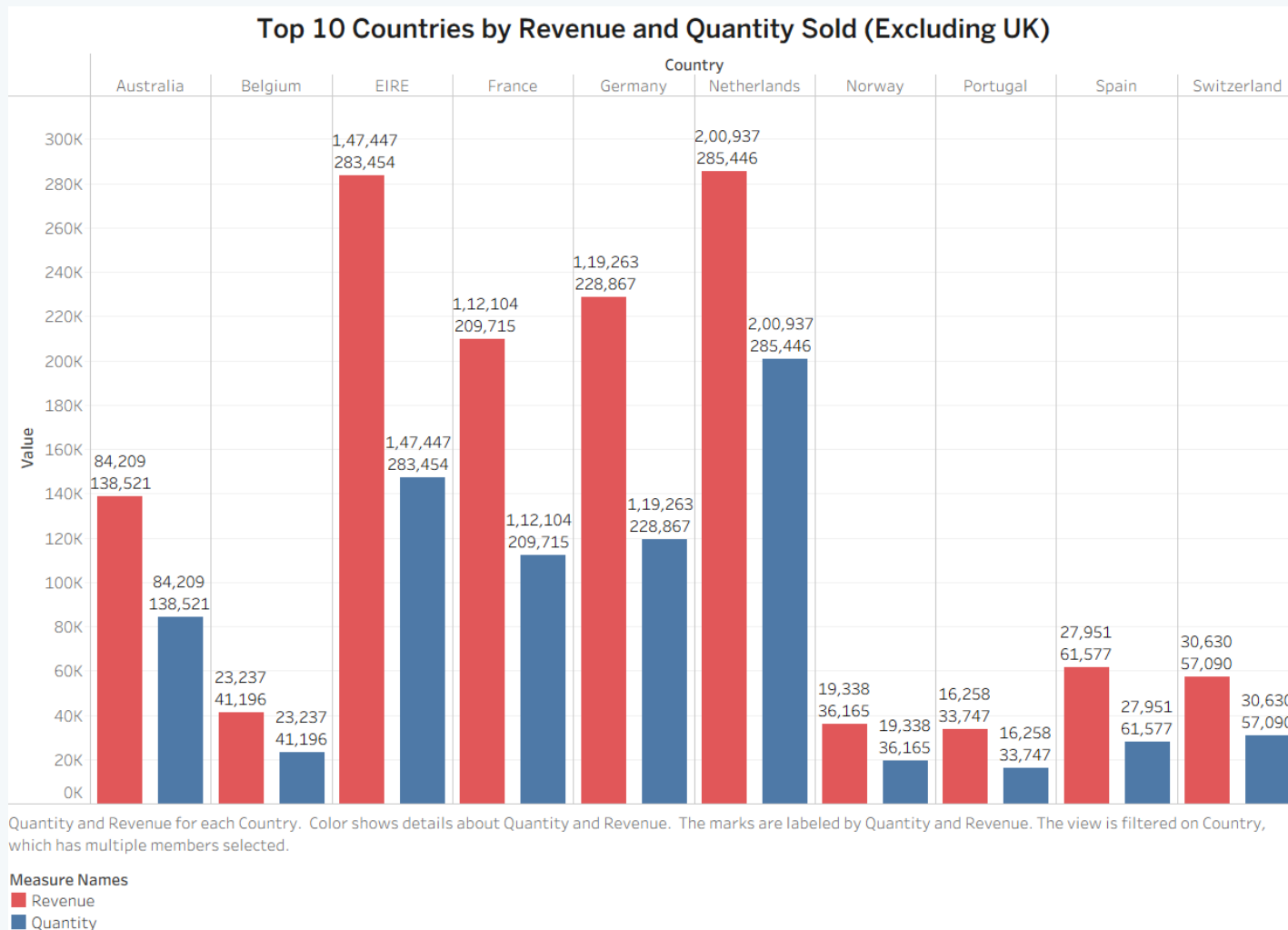
KEY INSIGHT

- Revenue held steady between \$520K–\$770K from January through August.
- A sharp climb begins in September, peaking in November at \$1.51M — nearly double the year's typical monthly revenue.
- December's drop to \$639K is not a real decline — the dataset only covers data through December 9th.

Q2

Top 10 Countries — Revenue & Quantity

Asked by: CMO • UK excluded



THE QUESTION

Identify the top 10 revenue-generating countries (excluding the UK), and show quantity sold alongside revenue for each.

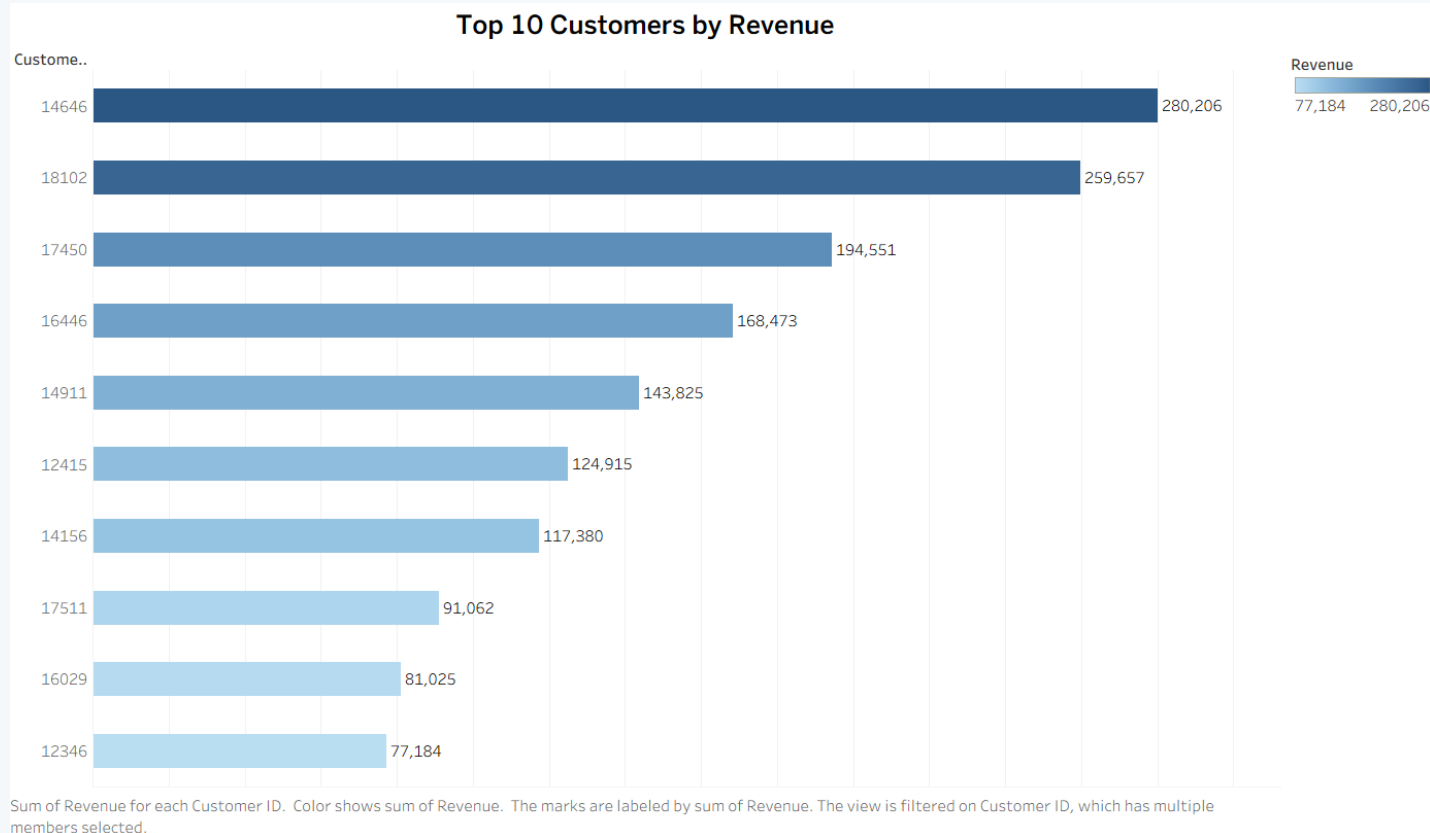
KEY INSIGHT

- Netherlands (\$285K) and EIRE (\$283K) are nearly tied for the top spot — together with Germany and France, they form a clear top-4 cluster.
- These four countries drive the large majority of non-UK revenue; all other countries trail well behind.
- Revenue and quantity rise and fall together for every country — no market shows a mismatch between volume and earnings, suggesting consistent pricing across regions.

Q3

Top 10 Customers by Revenue

Asked by: CMO



THE QUESTION

Show the top 10 customers by revenue, ranked from highest to lowest, to identify who to prioritize for retention.

KEY INSIGHT

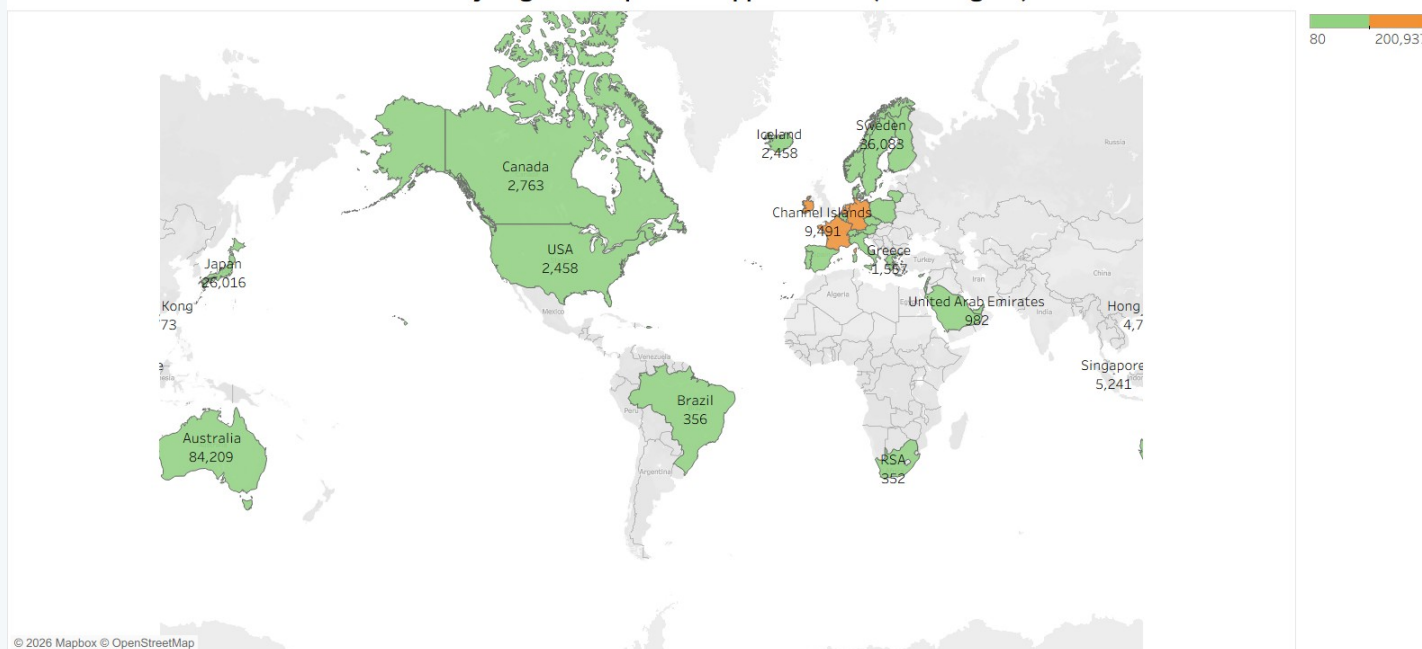
- Customers 14646 (\$280K) and 18102 (\$260K) stand far above the rest — together they out-earn the next three customers combined.
- Revenue declines steadily after the top two, down to \$77K for the 10th-ranked customer.
- This concentration is a retention risk: losing even one top customer would meaningfully impact total revenue.

Q4

Product Demand by Region

Asked by: CEO • UK excluded

Product Demand by Region — Expansion Opportunities (Excluding UK)



© 2026 Mapbox © OpenStreetMap

Map based on Longitude (generated) and Latitude (generated). Color shows sum of Quantity. The marks are labeled by Country and sum of Quantity. Details are shown for Country. The view is filtered on Country and sum of Quantity. The Country filter keeps 37 members. The sum of Quantity filter includes everything.

THE QUESTION

View product demand across all countries in a single glance, to identify regions with strong expansion potential.

KEY INSIGHT

- Beyond the established European base, Sweden, the Channel Islands, and Australia show standout demand among non-core markets.
- Most other regions — USA, Canada, Japan, Brazil — currently show comparatively low demand.
- These lower-demand regions represent potential white space worth testing with focused marketing or logistics investment.

Recommendations for the CEO & CMO

What the data suggests for next year's strategy

1

Prepare early for the seasonal peak

Build inventory and launch marketing campaigns starting in August, ahead of the September–November revenue surge seen this year.

2

Protect and grow the core four markets

Netherlands, EIRE, Germany, and France drive the bulk of international revenue — defend this base while piloting expansion in Sweden, the Channel Islands, and Australia.

3

Reduce customer concentration risk

Set up a dedicated key-account program for top customers like 14646 and 18102, while actively working to diversify and grow the broader customer base.

4

Improve Customer ID capture at checkout

A meaningful share of transactions are missing a Customer ID, limiting customer-level analysis. Fixing this unlocks deeper segmentation and loyalty insights.

5

Extend this analysis going forward

Repeat this study with a full additional year of data to confirm the seasonal pattern holds, and build a forecasting model for next year's inventory planning.

Thank You

Questions & discussion welcome

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